

# Information, Martingales, and Multi-Period Markets

Financial Engineering · Section 2

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# Outline

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- 2.1 Information and filtrations
- 2.2 Conditional expectation
- 2.3 Martingales in discrete time
- 2.4 Multi-period markets and the FTAP

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# Information and Filtrations

## What is known when

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- The economic constraint is simple. A trading decision made at date  $t$  may use only information available at date  $t$ .
- A state  $\omega$  is now a *complete path* of the world through all dates  $t = 0, 1, \dots, T$ .
- At an intermediate date the path is only partly revealed. We know which *group* of states is still possible.
- We do not know which single state will occur.
- Those groups carry the information available at date  $t$ .

# Events, as a list of questions

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## Definition 2.1 (Events and $\sigma$ -algebras)

$\mathcal{G}$ , a collection of subsets of  $\Omega$ , is a  $\sigma$ -algebra on  $\Omega$  if

- (i)  $\Omega \in \mathcal{G}$ ;
- (ii)  $A \in \mathcal{G} \implies A^c \in \mathcal{G}$ ;
- (iii)  $A_1, A_2, \dots \in \mathcal{G} \implies \bigcup_{n \geq 1} A_n \in \mathcal{G}$ .

Members are *events*;  $\sigma(C)$  is the smallest  $\sigma$ -algebra containing  $C$ .

- $A \in \mathcal{G}$  means “did  $A$  happen?” can be answered.
- In  $(\Omega, \mathcal{F}, \mathbb{P})$ , one  $\omega$  is a **complete path** from date 0 to  $T$ , not a local outcome.

## Is it a $\sigma$ -algebra?

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### Example 2.2

Let  $\Omega = \{1, 2, 3, 4\}$ . (a) Which of these is a  $\sigma$ -algebra on  $\Omega$ ?

$$\mathcal{G}_1 = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\}, \quad \mathcal{G}_2 = \{\emptyset, \{1\}, \{1, 2\}, \Omega\}.$$

(b) Write out  $\sigma(C)$  for  $C = \{\{1\}, \{2\}\}$ .

# Filtrations

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## Definition 2.3 (Filtration)

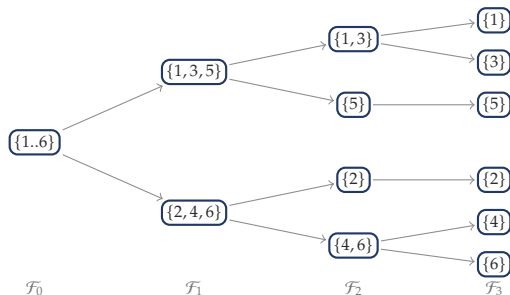
A *filtration* is a family  $\mathbb{F} = (\mathcal{F}_t)_{t=0}^T$  of sub- $\sigma$ -algebras of  $\mathcal{F}$  with  $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$  whenever  $s \leq t$ . If  $A \in \mathcal{F}_t$ , an observer can say by date  $t$  whether  $A$  has occurred.

- Information grows: a question answerable yesterday is still answerable today.
- For the event-tree models we also assume  $\mathcal{F}_0$  trivial and  $\mathcal{F}_T = \mathcal{F}$ . Those are modelling assumptions, not part of the definition.
- On a finite  $\Omega$ ,  $\mathcal{F}_t$  groups the paths it cannot yet tell apart. Those groups are the **cells**, or atoms, of  $\mathcal{F}_t$ .

# A die revealed in three stages

## Example 2.4

A die is rolled once,  $\Omega = \{1, \dots, 6\}$ . Date 1: the parity. Date 2: also whether it is  $\leq 3$  or  $\geq 4$ . Date 3: the number.



## What the die filtration shows

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- Cells refine:  $\{1, 3, 5\} = \{1, 3\} \cup \{5\}$  and  $\{2, 4, 6\} = \{2\} \cup \{4, 6\}$ .
- Events per date: 2, 4, 16, 64, since a date with  $k$  cells has  $2^k$  events.
- Date-2 cells are **not** all the same size.
- On  $\{5\}$  and  $\{2\}$  the number is already known at date 2. Information need not arrive at the same rate along every path.

# Measurable, and adapted

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## Definition 2.5 (Measurable, adapted)

(i)  $Z$  is  $\mathcal{F}_t$ -measurable if

$$\{Z \in B\} \in \mathcal{F}_t \quad \text{for every interval } B \subseteq \mathbb{R}.$$

(ii)  $X = (X_t)_{t=0}^T$  is *adapted* to  $\mathbb{F}$  if every  $X_t$  is  $\mathcal{F}_t$ -measurable.

- Every question about  $Z$  can be answered at date  $t$ . On a finite  $\Omega$ : constant on every date- $t$  cell.
- A random variable is  $\mathcal{F}_t$ -measurable; a process is adapted to  $\mathbb{F}$ . Prices and strategies are adapted.

## Which variables are $\mathcal{F}_2$ -measurable?

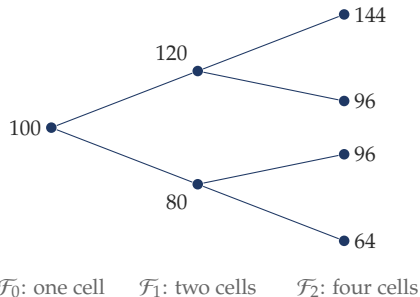
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### Example 2.6

Same filtration,  $X$  the number rolled. Test  $X$ ;  $Y = \mathbf{1}\{X \text{ is even}\}$ ; and  $W$ , the smallest number still possible at date 2. Then name one event in  $\mathcal{F}_2$  but not in  $\mathcal{F}_1$ .

## An event tree

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- Prices:  $S_0 = 100$ ,  $u = 1.2$ ,  $d = 0.8$ ; Example 2.11 uses them again.
- At date 1 you know only which of the two cells you are in.
- $ud$  and  $du$  share the price 96 but are different states.



# Conditional Expectation

# Conditional expectation

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## Definition 2.7 (Conditional expectation)

For integrable  $X$ , a random variable  $Y$  is a version of  $\mathbb{E}[X \mid \mathcal{F}_t]$  if

- (i)  $Y$  is  $\mathcal{F}_t$ -measurable; and
  - (ii)  $\mathbb{E}[\mathbf{1}_A Y] = \mathbb{E}[\mathbf{1}_A X]$  for every  $A \in \mathcal{F}_t$ .
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- (i): the forecast may use only date- $t$  information.
  - (ii): on every event already decidable at  $t$ , the forecast keeps the average of  $X$ .
  - Not a number: a *random variable*, and  $\mathcal{F}_t$ -measurable.

## Computing it on a finite tree

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- Write  $\pi(C)$  for the probability of a cell. On each date- $t$  cell,

$$\mathbb{E}[X \mid C] = \frac{\sum_{\omega \in C} \pi_{\omega} X(\omega)}{\pi(C)}.$$

- This is the **computation rule**, not the definition. Every calculation in this section uses it.
- It meets both conditions: constant on each cell, and correct on average over every union of cells.

## More information, sharper forecast

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### Example 2.8

The die filtration of Example 2.4,  $X$  the number rolled. Compute  $\mathbb{E}[X \mid \mathcal{F}_1]$  and  $\mathbb{E}[X \mid \mathcal{F}_2]$ , check the tower property, and check both average to  $\mathbb{E}[X]$ .

- Date 1:  $\frac{1+3+5}{3} = 3$  on  $\{1, 3, 5\}$ ,  $\frac{2+4+6}{3} = 4$  on  $\{2, 4, 6\}$ .
- Date 2: 2, 5, 2, 5 on  $\{1, 3\}$ ,  $\{5\}$ ,  $\{2\}$ ,  $\{4, 6\}$ . More information, a forecast that moves further.
- Tower on  $\{1, 3, 5\}$ :  $\frac{2}{3}(2) + \frac{1}{3}(5) = 3$ ; on  $\{2, 4, 6\}$ :  $\frac{1}{3}(2) + \frac{2}{3}(5) = 4$ .
- Both average to  $3.5 = \mathbb{E}[X]$ . Proposition 2.9 says why.

# The three properties we use constantly

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## Proposition 2.9 (Properties of conditional expectation)

For integrable  $X, Y$ ,  $s \leq t$ , and bounded  $\mathcal{F}_t$ -measurable  $Z$ :

- **(a) Linearity:**  $\mathbb{E}[aX + bY \mid \mathcal{F}_t] = a\mathbb{E}[X \mid \mathcal{F}_t] + b\mathbb{E}[Y \mid \mathcal{F}_t]$ .
- **(b) Taking out what is known:**  $\mathbb{E}[ZX \mid \mathcal{F}_t] = Z\mathbb{E}[X \mid \mathcal{F}_t]$ .
- **(c) Tower property:**  $\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_t] \mid \mathcal{F}_s] = \mathbb{E}[X \mid \mathcal{F}_s]$ , and in particular  $\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_t]] = \mathbb{E}[X]$ .
- Forecasts respect sums and multiples. A factor already known at date  $t$  can be moved outside the forecast.
- A forecast of a later forecast is just the earlier forecast.

## Where this is going

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- Soon  $X$  will be the future value of a derivative.
- Its value today will be a forecast of that future value, under a reweighted measure written  $Q$ :

$$V_t = \frac{1}{1+r} \mathbb{E}^Q[ V_{t+1} \mid \mathcal{F}_t ].$$

- Nothing so far justifies this. Here  $r$  is the one-period riskless rate, and  $Q$  is defined only in §2.4 (Definition 2.17).
- Everything between here and there is built to make it true.

## When the states are not equally likely

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### Example 2.10

Four states with  $\pi = (0.4, 0.2, 0.3, 0.1)$ ; the date-1 cells are  $A = \{\omega_1, \omega_2\}$  and  $B = \{\omega_3, \omega_4\}$ . Let  $X = (10, 4, 8, 2)$  and  $Z = (3, 3, 5, 5)$ . (a) Find  $\mathbb{E}[X \mid \mathcal{F}_1]$  and check the tower property. (b) Check  $\mathbb{E}[ZX \mid \mathcal{F}_1] = Z \mathbb{E}[X \mid \mathcal{F}_1]$ . (c) Repeat (b) with  $Z' = (3, 5, 3, 5)$  and say what goes wrong.

## Why the definition breaks in continuous time

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- In Section 3 the states are continuous paths, and  $\mathcal{F}_t$  records the path seen so far.
- The cell containing the true state is then every path agreeing with what has been seen.
- That set has **probability zero**.
- The cell probability is zero, so the tree formula reads 0/0 and computes nothing.

## Why the definition survives

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- Definition 2.7 never divides.
- Measurable, plus correct averages over every event: that already pins the forecast down.
- Definition 2.5 goes the same way: “constant on every cell” fails, the Borel condition does not.
- So the characterization is the definition, and the tree formula is only a convenience.

## Back to the two-period tree

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### Example 2.11

On the two-period tree let  $S_0 = 100$ ,  $u = 1.2$ ,  $d = 0.8$  and  $r = 0$ , so  $S_2 \in \{144, 96, 96, 64\}$ , with physical up-probability  $p = 0.6$  at each date, independently. Compute  $\mathbb{E}[S_2 \mid \mathcal{F}_1]$  and verify the tower property  $\mathbb{E}[\mathbb{E}[S_2 \mid \mathcal{F}_1]] = \mathbb{E}[S_2]$ .

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# Martingales in Discrete Time

# Martingales

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## Definition 2.12 (Martingale)

An adapted, integrable process  $X$  is a *martingale* for a given filtration and measure if for every  $t$ ,

$$\mathbb{E}[X_{t+1} \mid \mathcal{F}_t] = X_t.$$

With  $\leq$  it is a *supermartingale*; with  $\geq$ , a *submartingale*.

- Given everything known today, tomorrow's expected value equals today's value. The game is fair.
- A supermartingale drifts down, so it is an unfavorable game. A submartingale drifts up.

## A fair game and an unfair one

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### Example 2.13

A gambler bets \$1 on each toss of a coin, heads with probability  $p$ , independently. Let  $X_t$  be cumulative winnings after  $t$  tosses,  $X_0 = 0$ .

- $\mathbb{E}[X_{t+1} \mid \mathcal{F}_t] = X_t + p(+1) + (1-p)(-1) = X_t + (2p - 1)$ .
- $p = \frac{1}{2}$ : added term 0, a **martingale**. The game is fair.
- $p = 0.6$ : added term  $0.2 > 0$ , a **submartingale**.  $p = 0.4$ :  $-0.2 < 0$ , a **supermartingale**.
- Unconditionally  $\mathbb{E}[X_t] = (2p - 1)t$ . Only the fair game leaves the mean where it started.

## Martingale with respect to *what*

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- Whether a process is a martingale depends on **both** the filtration and the measure. That flexibility is the whole point.
- The stock of Example 2.11 is *not* a martingale under the physical measure, because  $\mathbb{E}[S_{t+1} \mid \mathcal{F}_t] = 1.04 S_t > S_t$ : a positive expected excess return under  $\mathbb{P}$ . In an equilibrium reading, that is the risk premium investors require.
- It becomes a martingale when the probabilities are reweighted to  $q = \frac{1}{2}$ , since  $\frac{1}{2}(1.2) + \frac{1}{2}(0.8) = 1$ .
- Reweighting probabilities to remove drift is what the risk-neutral measure of Section 1 did in one period.

# Constant expectation

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## Proposition 2.14

A martingale has constant expectation:  $\mathbb{E}[X_t] = \mathbb{E}[X_0]$  for all  $t$ . If  $\mathcal{F}_0$  is trivial,  $X_0$  is constant and  $\mathbb{E}[X_t] = X_0$ .

- A fair game creates no expected gain at any horizon.
- The converse fails. Constant expectation at each date does not make the increments conditionally fair.

# Every forecast process is a martingale

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## Example 2.15

The die filtration of Example 2.4, with  $X$  the number rolled. Put  $M_t = \mathbb{E}[X \mid \mathcal{F}_t]$  for  $t = 0, 1, 2, 3$ . Show that  $M$  is a martingale.

- $M_0 = 3.5$ ;  $M_1 = 3, 4$ ;  $M_2 = 2, 5, 2, 5$ ;  $M_3 = X$ .
- $\mathbb{E}[M_1] = \frac{1}{2}(3) + \frac{1}{2}(4) = 3.5 = M_0$ ; on  $\{1, 3, 5\}$ ,  
 $\mathbb{E}[M_2 \mid \{1, 3, 5\}] = \frac{2}{3}(2) + \frac{1}{3}(5) = 3 = M_1$ .
- Nothing depends on the die:  
 $\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_{t+1}] \mid \mathcal{F}_t] = \mathbb{E}[X \mid \mathcal{F}_t] = M_t$  by the tower property.
- So every random variable generates a martingale this way.

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# Multi-Period Markets and the FTAP

## Trading through time

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- $S_t$  is the vector of risky prices, adapted; the riskless account is  $B_t = (1 + r)^t$ .
- A *trading strategy*  $\theta_t$  is the portfolio chosen at date  $t$ , after  $\mathcal{F}_t$  is observed, and held into  $t + 1$ .
- So  $\theta_t$  is  $\mathcal{F}_t$ -measurable. You choose today's position with today's information and nothing later.

## Adapted or predictable?

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- Many texts index holdings by the period they are held over:  $\theta_t$  is the portfolio held from  $t - 1$  to  $t$ , so  $\theta_t$  is  $\mathcal{F}_{t-1}$ -measurable.
- Such a process is adapted too, since  $\mathcal{F}_{t-1} \subseteq \mathcal{F}_t$ . But it is **stronger** than adapted, and the stronger property has its own name: **predictable**.
- In that indexing the slogan is: **prices are adapted, trading strategies are predictable**. Continuous-time stochastic integration turns on it.
- We index the other way, so the same restriction reads  $\theta_t$  is  $\mathcal{F}_t$ -measurable. No question about when  $\theta_0$  was chosen.

# Self-financing

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## Definition 2.16 (Self-financing strategy)

All rebalancing is done at market prices, with no cash added or withdrawn: in every state,

$$V_t = \theta_{t-1}^\top \bar{S}_t = \theta_t^\top \bar{S}_t \quad (t = 1, \dots, T-1),$$

where  $\bar{S}_t$  stacks the risky prices and the riskless account.

- The value carried into a date equals the cost of the portfolio carried out of it.

## Arbitrage, over many dates

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- An *arbitrage* is a self-financing strategy with  $V_0 \leq 0$  and  $V_T \geq 0$  in every state, with **at least one strict inequality**: either  $V_0 < 0$ , or  $V_T > 0$  in some state.
- Those are exactly the two types of Definition 1.3.
- **Type I:**  $V_0 = 0$  with  $V_T > 0$  somewhere. Nothing paid today, something received later.
- **Type II:**  $V_0 < 0$  with  $V_T \geq 0$ . Money received today, nothing owed later.

# Equivalent martingale measure

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## Definition 2.17 (EMM)

$Q$  is an *equivalent martingale measure* if (i)  $Q$  is *equivalent* to  $\mathbb{P}$ , written  $Q \sim \mathbb{P}$ :  $Q(A) = 0$  if and only if  $\mathbb{P}(A) = 0$ ; and (ii) discounted prices are  $Q$ -martingales,  $\mathbb{E}^Q[S_{t+1}/B_{t+1} \mid \mathcal{F}_t] = S_t/B_t$ .

- Every physical state here has  $\pi_\omega > 0$ , so on this finite space (i) says exactly  $Q(\{\omega\}) > 0$  for every  $\omega$ .
- $Q$  may reweight the states. It may not declare a possible state impossible.

# The fundamental theorem, multi-period

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## Theorem 2.18 (FTAP, multi-period)

A finite multi-period market admits no arbitrage **if and only if** an equivalent martingale measure exists.

- The reading is the one from Section 1. No free lunch means the same thing as a consistent system of positive “prices of probability”.
- The only new feature is where that consistency has to hold. It must now hold *conditionally at every node* of the tree.

## Where this leaves us

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- $\mathcal{F}_t$  says what is known at date  $t$ .
- $\mathbb{E}[\cdot | \mathcal{F}_t]$  forecasts with that information.
- A martingale is a process whose forecast equals its current value.
- The FTAP: no arbitrage  $\iff$  an equivalent  $Q$  exists making discounted prices martingales.
- Section 5 works the binomial case out in full, as a numerical method.